

Dynamic Systems Development Method (DSDM)

1. Introduction

DSDM (Dynamic Systems Development Method) is an Agile software development method. It was developed to provide a more structured approach to rapid software development while still allowing flexibility and frequent user feedback.

It focuses on:

- Rapid delivery
- Active user involvement
- Iterative and incremental development
- Fixed time and cost
- High-quality software
- Continuous communication and collaboration

Simple Definition

DSDM is an Agile method that develops software in small increments through continuous user involvement, iterative development, timeboxing, and requirement prioritization.

2. Why Was DSDM Introduced?

Traditional development models such as Waterfall have a fixed sequence:

Requirements → Design → Coding → Testing → Deployment

The problem is that requirements may change during development. Making changes late can be expensive and time-consuming.

DSDM was introduced to solve this problem by:

- Developing the most important features first.
- Involving users throughout development.
- Delivering software in small increments.
- Accepting changes where possible.
- Keeping time and cost under control.

3. Main Idea of DSDM

The central idea of DSDM is:

Fix the time and cost, but prioritize and adjust the features.

For example, suppose a project must be completed in 6 months with a fixed budget.

Instead of extending the deadline when new requirements appear:

- Essential features → Must be completed
- Important features → Completed if possible
- Less important features → Can be postponed

This helps ensure that a working product is delivered on time.

4. Key Characteristics of DSDM

1. Active User Involvement

Users and customers participate throughout the project.

They:

- Provide requirements.
- Review prototypes.
- Give feedback.
- Help prioritize features.
- Test the developed system.

Example:

While developing a College Management System, college staff regularly review the attendance and fee modules and suggest improvements.

2. Iterative Development

The software is not developed only once from beginning to end.

Instead, the development cycle is repeated:

Build → Test → Get Feedback → Improve → Build Again

Each iteration improves the software.

3. Incremental Development

The complete system is divided into smaller parts called increments.

For example, in an Online Shopping Application:

Increment 1

- Registration
- Login

Increment 2

- Product Catalog
- Search

Increment 3

- Shopping Cart

Increment 4

- Payment

Each increment adds new functionality to the system.

4. Timeboxing

Timeboxing is one of the most important concepts in DSDM.

A fixed amount of time is allocated to complete a particular task or iteration.

Example:

Activity	Fixed Time
Requirement discussion	1 week
Prototype development	2 weeks
Testing	1 week
Deployment preparation	1 week

The time period is generally not extended easily. If some less important features cannot be completed, they are postponed.

Simple Meaning:

Time is fixed, but the amount of work may be adjusted according to priority.

5. Requirement Prioritization

DSDM uses the MoSCoW technique to prioritize requirements.

M — Must Have

Essential features without which the system cannot work.

Example:

A banking application must allow users to securely log in.

S — Should Have

Important features but the system can still operate temporarily without them.

Example:

Downloadable transaction statements.

C — Could Have

Desirable features that can be included if time and resources are available.

Example:
Dark mode.

W — Won't Have This Time

Features that are not included in the current release but may be developed later.

Example:
AI-based financial advice.

5. DSDM Principles

DSDM is based on eight main principles.

1. Focus on the Business Need

Every feature should provide value to the business or user.

The team should first develop the features that are most important.

Example:

For a food delivery application, ordering food is more important than adding multiple app themes.

2. Deliver on Time

The project should meet its agreed deadlines.

Timeboxes help the team control the schedule.

3. Collaborate

Developers, customers, managers, and other stakeholders should work closely together.

Good communication reduces misunderstandings.

4. Never Compromise Quality

Speed should not reduce software quality.

The team should define quality standards and maintain them throughout development.

5. Build Incrementally from Firm Foundations

The system should be developed step by step on a stable foundation.

Important architecture and business requirements should be understood before adding new features.

6. Develop Iteratively

The software is continuously improved through feedback.

Develop → Review → Improve → Repeat

7. Demonstrate Control

The project team should regularly monitor:

- Progress
- Cost
- Time
- Quality
- Risks

This helps ensure that the project remains under control.

8. Communicate Continuously and Clearly

Continuous communication between team members and stakeholders is essential.

Problems should be identified and solved quickly.

6. DSDM Project Lifecycle

The DSDM lifecycle consists of several phases.

Phase 1: Pre-Project

Purpose:

Ensure that only suitable projects are started.

Activities:

- Identify the business problem.
- Identify project objectives.
- Check whether DSDM is appropriate.
- Define initial project scope.

Example:

A company decides that it needs a Customer Support Portal.

Phase 2: Feasibility

Purpose:

Determine whether the project is feasible.

The team checks:

- Technical feasibility
- Financial feasibility
- Resource availability
- Time availability
- Major risks

Output:

A decision about whether to continue with the project.

Phase 3: Foundations

Purpose:

Create a strong foundation before detailed development begins.

Activities:

- Understand business requirements.
- Define system architecture.
- Identify major risks.
- Prioritize requirements using MoSCoW.
- Create an initial development plan.

Output:

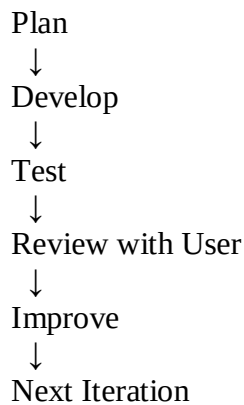
A clear foundation for development.

Phase 4: Evolutionary Development

This is the main development phase.

The software is developed through repeated iterations.

Cycle:



Each iteration produces a working part of the software.

Example:

Iteration 1: Login System
Iteration 2: User Dashboard
Iteration 3: Payment System
Iteration 4: Reports and Notifications

Phase 5: Deployment

The completed software is prepared for real users.

Activities:

- Final testing
- User Acceptance Testing (UAT)
- Data migration
- User training
- Deployment

The system is then released.

Phase 6: Post-Project

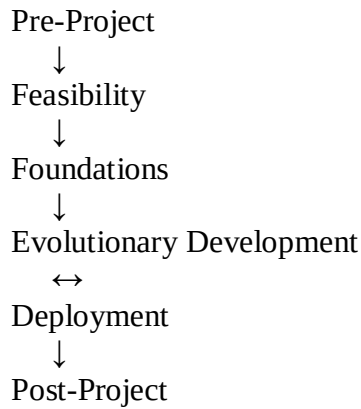
After deployment, the software is monitored and maintained.

Activities:

- Fix bugs
- Provide support
- Monitor performance

- Plan future releases
 - Add postponed features
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7. DSDM Lifecycle Diagram



Evolutionary Development may involve several iterations and continuous feedback before deployment.

8. Roles in DSDM

DSDM involves different stakeholders.

Business Sponsor

- Provides funding.
 - Supports the project.
 - Ensures business benefits.
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Business Visionary

- Defines the overall business vision.
 - Ensures that development supports business goals.
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Business Ambassador

- Represents the users.
 - Provides regular feedback to the development team.
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Project Manager

- Manages time, cost, resources, and risks.
 - Ensures that the project remains on track.
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Technical Coordinator

- Provides technical guidance.
 - Ensures that the technical solution is appropriate.
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Team Leader

- Coordinates the development team.
 - Helps organize day-to-day activities.
-

Developers

- Design and build the software.
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Testers

- Test the software.
 - Ensure quality requirements are met.
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9. Advantages of DSDM

1. Faster Delivery

Timeboxing helps deliver software within a fixed schedule.

2. Active User Participation

Continuous feedback increases customer satisfaction.

3. Flexible Requirements

Requirements can be changed and reprioritized.

4. Better Business Focus

Important features are developed first.

5. Controlled Cost and Time

The project focuses on fixed resources and deadlines.

6. Improved Quality

Testing and feedback occur throughout development.

7. Reduced Risk

Problems can be identified early.

10. Disadvantages of DSDM

1. Requires Active User Involvement

The method may fail if users are unavailable.

2. Requires Skilled Team Members

Team members need experience and strong communication skills.

3. Difficult for Very Large Projects

Coordination can become difficult.

4. Requirement Prioritization Can Be Difficult

Stakeholders may disagree about which features are most important.

5. Not Suitable for Every Project

Projects requiring completely fixed requirements and strict processes may need another model.

11. Example: Online Food Delivery Application

Suppose a company wants to develop a food delivery application in 4 months.

Step 1: MoSCoW Prioritization

Must Have

- User registration
- Restaurant listing
- Food ordering

- Payment

Should Have

- Order tracking
- Customer reviews

Could Have

- Dark mode
- Personalized recommendations

Won't Have This Time

- AI-based meal planning

Step 2: Timeboxing

Iteration	Duration	Features
Iteration 1	2 weeks	Registration and Login
Iteration 2	3 weeks	Restaurant and Menu
Iteration 3	3 weeks	Cart and Ordering
Iteration 4	2 weeks	Payment
Iteration 5	2 weeks	Tracking and Reviews

After every iteration, users provide feedback.

If time becomes limited, Could Have features are postponed, but Must Have features are completed.

This ensures a useful working application is delivered on time.

12. DSDM vs Scrum

DSDM	Scrum
Agile method	Agile framework
Strong focus on time and cost	Strong focus on Sprint delivery
Uses MoSCoW prioritization	Uses Product Backlog prioritization
Uses defined business and technical roles	Uses Product Owner, Scrum Master, and Developers
More structured governance	Simpler framework
Focuses strongly on business value	Focuses strongly on iterative product development

13. DSDM vs RAD

DSDM	RAD
Structured Agile method	Rapid development approach/process model
Uses timeboxing	Focuses mainly on rapid development
Uses MoSCoW prioritization	Often uses rapid prototyping
Strong project governance	Less structured
Focuses on business value and control	Focuses on speed and quick prototyping

Important Difference to Remember

RAD:

"Build software quickly."

DSDM:

"Build the most important software features quickly, within fixed time and cost."

Exam Answer Summary

Dynamic Systems Development Method (DSDM) is an Agile software development method that emphasizes rapid and iterative development, active user involvement, timeboxing, incremental delivery, and prioritization of requirements using the MoSCoW technique.

Key Features

- Timeboxing
- MoSCoW prioritization
- Active user involvement
- Iterative development
- Incremental delivery
- Continuous testing
- Strong business focus

DSDM Phases

Pre-Project → Feasibility → Foundations → Evolutionary Development → Deployment → Post-Project

One-Line Memory Formula

DSDM = Agile + User Involvement + Timeboxing + MoSCoW + Iterative Development + Fast Delivery

